



TG-500 Series Technical Datasheet

The TG-500 series is a next-generation edge computing gateway developed by TesPro for industrial IoT scenarios. Based on the MT798X (Wi-Fi 7 BE5000) core module, it is equipped with an ARM Cortex-A53 dual-core processor and a hardware network acceleration engine, and features an embedded Linux/Open-WRT operating system. It combines powerful computing power with high reliability and is suitable for harsh industrial environments.

- Supports 2G/3G/4G/5G multi-band cellular access and is compatible with major global operator network standards
- Supports multi-protocol access including IEC 61850/MMS, Modbus TCP/RTU, etc.
- Wi-Fi 7 high-speed wireless backhaul + Ethernet multi-port access
- Supports access via protocols such as IEC 61850/MMS and Modbus TCP/RTU
- Compatible with mainstream PLC brands such as Siemens, Mitsubishi, and Omron, and supports the OPC UA protocol
- Supports EMS data access for energy storage systems
- 12 – 36 V DC wide-range power input

Technical Specifications

Items	TG-524	TG-525
Version	4G	5G
Power Supply	12 – 36 V DC	12 – 36 V DC
Indicator Light	Power / Network / Communication	Power / Network / Communication
Region / Carrier	Global	Global
Reference Dimensions	125mm*108*50mm	125mm*108*50mm
CPU	RM Cortex-A53 dual-core, clocked at 1.6GHz, integrated hardware network acceleration engine	RM Cortex-A53 dual-core, clocked at 1.6GHz, integrated hardware network acceleration engine
Operating system	Open-WRT, supports customer C/C++ program running and secondary development	Open-WRT, supports customer C/C++ program running and secondary development
Memory	2GB DDR4	2GB DDR4
EMMC	8G/16G/32GB/64GB eMMC, standard 32G	8G/16G/32GB/64GB eMMC, standard 32G
FLASH	256MB/512MB NAND FLASH	256MB/512MB NAND FLASH
LTE-FDD	B1/ 2/ 3/ 4/ 5/ 7/ 8/ 12/ 13/ 18/ 19/ 20/ 25/ 26/ 28	B1/ 2/ 3/ 4/ 5/ 7/ 8/ 12/ 13/ 18/ 19/ 20/ 25/ 26/ 28
LTE-TDD	B38/ 39/ 40/ 41	B38/ 39/ 40/ 41
WCDMA	B1/ 2/ 4/ 5/ 6/ 8/ 19	B1/ 2/ 4/ 5/ 6/ 8/ 19
GSM/EDGE	B2/ B3/ 5/ 8	B2/ B3/ 5/ 8
Cellular Network	4G Cat 1 4G Cat 4	5G Redcap 5G SA
GNSS	GPS, GLONASS, BDS, Galileo, QZSS (optional)	GPS, GLONASS, BDS, Galileo, QZSS (optional)
LTE-FDD	Maximum 10 Mbps (downlink) / 5 Mbps (uplink)	Maximum 10 Mbps (downlink) / 5 Mbps (uplink)
LTE-TDD	Maximum 8.96 Mbps (downlink) / 3.1 Mbps (uplink)	Maximum 8.96 Mbps (downlink) / 3.1 Mbps (uplink)
WCDMA	Maximum 384 kbps (downlink) / Maximum 384 kbps (uplink)	Maximum 384 kbps (downlink) / Maximum 384 kbps (uplink)
EDGE	Maximum 296 kbps (downlink) / Maximum 236.8 kbps (uplink)	Maximum 296 kbps (downlink) / Maximum 236.8 kbps (uplink)
GPRS	Maximum 107 kbps (downlink) / Maximum 85.6 kbps (uplink)	Maximum 107 kbps (downlink) / Maximum 85.6 kbps (uplink)
LTE-FDD	Class 3 (23 dBm ±2 dB)	Class 3 (23 dBm ±2 dB)
LTE-TDD	Class 3 (23 dBm ±2 dB)	Class 3 (23 dBm ±2 dB)
WCDMA	Class 3 (24 dBm +1/-3 dB)	Class 3 (24 dBm +1/-3 dB)
GSM/850	Class 4 (33 dBm ±2 dB)	Class 4 (33 dBm ±2 dB)
Number of serial ports	1 RS232, 2 RS485 (can work simultaneously)	1 RS232, 2 RS485 (can work simultaneously)
Serial port pins	RS232 – terminals (V/TX/RX/GND), RS232 – DB9 (2-RX/3-TX/5-GND), RS485 – terminals (A/B/G)	RS232 – terminals (V/TX/RX/GND), RS232 – DB9 (2-RX/3-TX/5-GND), RS485 – terminals (A/B/G)
Data bits	7, 8	7, 8
Stop bits	1, 2	1, 2
Parity bits	NONE, ODD, EVEN	NONE, ODD, EVEN
Baud rate	300-230400bps	300-230400bps
Bus Standard	ISO 11898	ISO 11898

Supported Baud Rate	5 kbps ~ 1 Mbps	5 kbps ~ 1 Mbps
Bus Type	Differential CANH/CANL 1.5-3.0V	Differential CANH/CANL 1.5-3.0V
Operating Modes	High-speed mode/Slope control mode/Standby low-power mode	High-speed mode/Slope control mode/Standby low-power mode
Networking Capability	Maximum number of connected nodes greater than 110	Maximum number of connected nodes greater than 110
Dual SIM Card Slots	Dual SIM	Dual SIM
Wi-Fi Antenna	2 Wi-Fi antennas, SMA external screw internal hole	2 Wi-Fi antennas, SMA external screw internal hole
4G Antenna	2 4G antennas, SMA external screw internal hole	2 4G antennas, SMA external screw internal hole
Button	Reset, restores default parameters	Reset, restores default parameters
2-P Socket	Power socket	Power socket
7-P Socket	2*RS485, 1*RS232	2*RS485, 1*RS232
4-P Socket	Independent 2*CAN, no interference	Independent 2*CAN, no interference
8-P Socket	2*DI, 2*DO	2*DI, 2*DO
TF Card	Micro SD Card	Micro SD Card
USB	USB 3.0	USB 3.0
Data Acquisition Methods	Real-time acquisition/reporting, interval acquisition/timed reporting, custom data acquisition methods	Real-time acquisition/reporting, interval acquisition/timed reporting, custom data acquisition methods
Meter Protocols	IEC1107/DLT645/DLMS	IEC1107/DLT645/DLMS
Modbus	Modbus RTU	Modbus RTU
Serial Port Protocols	Pass-through, TCP client/server, UDP, Modbus RTU gateway	Pass-through, TCP client/server, UDP, Modbus RTU gateway
Network Protocols	TCP, UDP, HTTP, HTTPS, etc.	TCP, UDP, HTTP, HTTPS, etc.
Industrial Protocols	Siemens/CAN/PROFINET/OPC/Ether-CAT/IEC61850, etc.	Siemens/CAN/PROFINET/OPC/Ether-CAT/IEC61850, etc.
Operating temperature	-40 ~ +75 °C	-40 ~ +75 °C
Operating humidity	10% ~ 95% (non-condensing)	10% ~ 95% (non-condensing)
Storage temperature	-20 ~ +80 °C	-20 ~ +80 °C
Storage humidity	10% ~ 95% (non-condensing)	10% ~ 95% (non-condensing)
Trust Zone	supported	supported
Secure Boot	supported	supported
TPM	Built-in TPM security chip, version TPM2.0, supports domestically produced domestic IT chips.	Built-in TPM security chip, version TPM2.0, supports domestically produced domestic IT chips.
Reverse polarity protection	supported	supported
Overvoltage & overcurrent protection	supported	supported
Lightning protection	supported	supported
Surge protection	supported	supported



Tespro Electronics Co., Ltd.

<https://www.tespro.com>

Email: info@tespro.com / sales@tespro.com

Mobile/whatsapp : China: +86-13928019921, HK: +852-46094648, US: +1-2136104797

We have offices in multiple countries: China, US, UK, AUS, HK. If you encounter an agent, please contact us first to confirm.